

CSSE 220 – Object-Oriented Software Development
Rose-Hulman Institute of Technology

Worksheet Tracing Practice

1. Draw the recursion trace for `multiplyDown(3)` and identify the final result.

```
1 /**
2  * Returns n * (n-1) * ... * 1
3  */
4  public int multiplyDown(int n) {
5      if (n <= 1) {
6          return 1;
7      }
8      return n * multiplyDown(n - 1);
9  }
```

2. Draw the recursion trace for `repeat("Hi", 3)` and identify the final returned string.

```
1 /**
2  * Repeats a word n times
3  */
4 public String repeat(String word, int n) {
5     if (n <= 0) {
6         return "";
7     }
8     return word + repeat(word, n - 1);
9 }
```